Fractions of a Group

Share the group into equal parts, then count the parts you need.

1/3 of 12: make 3 equal groups



$\frac{1}{3}$ of 12 = 4

One group out of 3.

Think division

$12 \div 3 = 4$

The bottom number tells how many groups.

2/3 of 12: two of the groups



$\frac{2}{3}$ of 12 = 8

4 + 4 = 8, two groups.

The steps

1. Divide by the bottom number: that's one group.
2. Times the top number: that many groups.

$\frac{3}{4}$ of 20: $20 \div 4 = 5$, $3 \times 5 = 15$

Your turn!



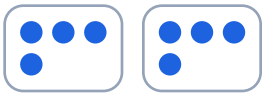
$\frac{1}{4}$ of 8 = ☐

Fractions of a Group: Practice

Bottom number: how many equal groups. Top number: how many groups to count.

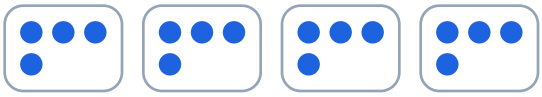
Part 1 One group The rings show the equal groups. How many in one group?

1)  $\frac{1}{3}$ of 15 = ☐

2)  $\frac{1}{2}$ of 8 = ☐

3)  $\frac{1}{5}$ of 20 = ☐

4)  $\frac{1}{6}$ of 18 = ☐

5)  $\frac{1}{4}$ of 16 = ☐

6)  $\frac{1}{6}$ of 12 = ☐

Part 2 Count the groups Count the dots in the marked groups.

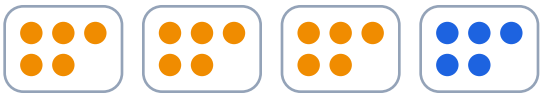
7)  $\frac{3}{4}$ of 12 = ☐

Fractions of a Group: Practice (continued)

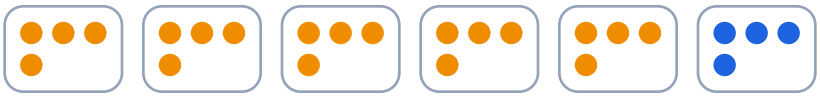
Part 2 Count the groups (continued)

8)  $\frac{2}{5}$ of 15 = ☐

9)  $\frac{2}{3}$ of 18 = ☐

10)  $\frac{3}{4}$ of 20 = ☐

11)  $\frac{4}{5}$ of 10 = ☐

12)  $\frac{5}{6}$ of 24 = ☐

Part 3 Make the groups Circle equal groups first. The bottom number says how many.

13)  $\frac{2}{3}$ of 9 = ☐

14)  $\frac{1}{4}$ of 12 = ☐

Fractions of a Group: Practice (continued)

Part 3 Make the groups (continued)

15)  $\frac{1}{2}$ of 10 = ☐

16)  $\frac{3}{5}$ of 15 = ☐

17)  $\frac{3}{4}$ of 16 = ☐

18)  $\frac{2}{7}$ of 14 = ☐

Part 4 Divide, then multiply One group first, then the groups you need.

19) 24 ÷ 4 = ☐ so $\frac{3}{4}$ of 24 = ☐

20) 30 ÷ 5 = ☐ so $\frac{2}{5}$ of 30 = ☐

21) 21 ÷ 3 = ☐ so $\frac{2}{3}$ of 21 = ☐

22) 32 ÷ 8 = ☐ so $\frac{3}{8}$ of 32 = ☐

23) 36 ÷ 6 = ☐ so $\frac{5}{6}$ of 36 = ☐

24) 28 ÷ 4 = ☐ so $\frac{3}{4}$ of 28 = ☐

25) 40 ÷ 10 = ☐ so $\frac{7}{10}$ of 40 = ☐

26) 27 ÷ 9 = ☐ so $\frac{4}{9}$ of 27 = ☐

Fractions of a Group: Practice (continued)

Part 5 One group No help this time.

27) $\frac{1}{4}$ of 20 = ☐

28) $\frac{1}{5}$ of 35 = ☐

29) $\frac{1}{3}$ of 18 = ☐

30) $\frac{1}{8}$ of 48 = ☐

31) $\frac{1}{6}$ of 30 = ☐

32) $\frac{1}{2}$ of 14 = ☐

33) $\frac{1}{10}$ of 60 = ☐

34) $\frac{1}{9}$ of 45 = ☐

35) $\frac{1}{7}$ of 28 = ☐

36) $\frac{1}{8}$ of 40 = ☐

37) $\frac{1}{3}$ of 24 = ☐

38) $\frac{1}{6}$ of 42 = ☐

Fractions of a Group: Practice (continued)

Part 6 More than one group No help this time.

39) $\frac{3}{5}$ of 20 = ☐

40) $\frac{5}{6}$ of 24 = ☐

41) $\frac{3}{4}$ of 16 = ☐

42) $\frac{5}{8}$ of 40 = ☐

43) $\frac{2}{3}$ of 30 = ☐

44) $\frac{9}{10}$ of 50 = ☐

45) $\frac{2}{9}$ of 36 = ☐

46) $\frac{4}{7}$ of 35 = ☐

47) $\frac{3}{4}$ of 32 = ☐

48) $\frac{5}{6}$ of 18 = ☐

49) $\frac{4}{5}$ of 45 = ☐

50) $\frac{3}{8}$ of 56 = ☐

Fractions of a Group: Practice (continued)

Part 7 Hours and dollars No help this time. 1 hour = 60 minutes, 1 dollar = 100 cents.

51) $\frac{1}{4}$ of an hour = ☐ minutes

52) $\frac{1}{2}$ of a dollar = ☐ cents

53) $\frac{3}{4}$ of an hour = ☐ minutes

54) $\frac{1}{4}$ of a dollar = ☐ cents

55) $\frac{1}{3}$ of an hour = ☐ minutes

56) $\frac{1}{10}$ of a dollar = ☐ cents

57) $\frac{2}{3}$ of an hour = ☐ minutes

58) $\frac{3}{4}$ of a dollar = ☐ cents

Part 8 A fraction of a group

59) $\frac{1}{2}$ of 10 = ☐

60) $\frac{1}{5}$ of 45 = ☐

61) $\frac{1}{4}$ of 36 = ☐

Fractions of a Group: Practice (continued)

Part 8 A fraction of a group (continued)

62) $\frac{1}{2}$ of 18 = ☐

63) $\frac{1}{5}$ of 30 = ☐

64) $\frac{1}{5}$ of 15 = ☐

65) $\frac{1}{8}$ of 24 = ☐

66) $\frac{1}{4}$ of 28 = ☐

67) $\frac{1}{6}$ of 24 = ☐

68) $\frac{1}{2}$ of 4 = ☐

69) $\frac{1}{8}$ of 64 = ☐

70) $\frac{1}{2}$ of 6 = ☐

71) $\frac{1}{6}$ of 12 = ☐

72) $\frac{1}{4}$ of 16 = ☐

73) $\frac{1}{3}$ of 9 = ☐

Fractions of a Group: Practice (continued)

Part 8 A fraction of a group (continued)

74) $\frac{1}{3}$ of 21 = ☐

75) $\frac{1}{8}$ of 16 = ☐

76) $\frac{1}{5}$ of 40 = ☐

77) $\frac{1}{2}$ of 8 = ☐

78) $\frac{1}{6}$ of 54 = ☐

79) $\frac{1}{3}$ of 15 = ☐

80) $\frac{1}{3}$ of 6 = ☐

81) $\frac{1}{8}$ of 32 = ☐

82) $\frac{1}{4}$ of 12 = ☐

Part 9 More than one part

83) $\frac{5}{6}$ of 42 = ☐

84) $\frac{4}{5}$ of 15 = ☐

Fractions of a Group: Practice (continued)

Part 9 More than one part (continued)

85) $\frac{5}{8}$ of 32 = ☐

86) $\frac{4}{6}$ of 30 = ☐

87) $\frac{3}{5}$ of 10 = ☐

88) $\frac{2}{3}$ of 18 = ☐

89) $\frac{4}{8}$ of 56 = ☐

90) $\frac{2}{3}$ of 9 = ☐

91) $\frac{3}{5}$ of 40 = ☐

92) $\frac{6}{8}$ of 56 = ☐

93) $\frac{2}{4}$ of 8 = ☐

94) $\frac{3}{4}$ of 8 = ☐

95) $\frac{5}{8}$ of 16 = ☐

96) $\frac{3}{4}$ of 12 = ☐

Fractions of a Group: Practice (continued)

Part 9 More than one part (continued)

97) $\frac{4}{5}$ of 10 = ☐

98) $\frac{2}{3}$ of 6 = ☐

99) $\frac{2}{3}$ of 12 = ☐

100) $\frac{2}{5}$ of 35 = ☐

Part 10 Remember? Problems from earlier packets.

101) $\frac{4}{10}$ ☐ $\frac{7}{10}$

102) $\frac{4}{10} = \frac{2}{☐}$

103) $\frac{15}{3}$ = ☐

104) Round 917 to the nearest hundred: ☐

105) 9 groups of 2 = ☐

106) $\frac{1}{6}$ ☐ $\frac{1}{10}$

Fractions of a Group: Practice (continued)

Part 11 Word problems Draw the groups.

107) There are 12 kids. 1/3 of them wear hats. How many kids wear hats?

Number sentence _____

Answer ☐ kids

Draw or show your work

108) Ana has 20 stickers. She gives away 1/4 of them. How many stickers does she give away?

Number sentence _____

Answer ☐ stickers

Draw or show your work

109) A box has 18 crayons. 2/3 of them are new. How many crayons are new?

Number sentence _____

Answer ☐ crayons

Draw or show your work

110) Leo has 30 marbles. 3/5 of them are blue. How many marbles are blue?

Number sentence _____

Answer ☐ marbles

Draw or show your work

111) Challenge: Mia read 3/4 of a 40-page book. How many pages does she have left to read?

Number sentence _____

Answer ☐ pages

Draw or show your work

Check-In: Fractions of a Group

Show what you know. No help this time. Score: _____ / 9

Part 1 Count the groups

1)  $\frac{3}{5}$ of 20 = ☐

Part 2 Make the groups

2)  $\frac{2}{4}$ of 12 = ☐

Part 3 One group

3) $\frac{1}{7}$ of 42 = ☐

4) $\frac{1}{6}$ of 36 = ☐

5) $\frac{1}{4}$ of 32 = ☐

Part 4 More than one group

6) $\frac{4}{7}$ of 21 = ☐

7) $\frac{2}{5}$ of 45 = ☐

8) $\frac{8}{10}$ of 20 = ☐

9) $\frac{7}{8}$ of 16 = ☐